5A · 6A · 10A · Atmosphere · EvolLand

Test 3 Master Test Review

Simplified · 38 Review Questions

TEACHER EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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TEACHER EDITION

Test 3 Master Test Review: Simplified

38 short review questions in subject order, matching the student edition question for question. Each item gives a brief answer and where students find it (linked). The QTI line gives the Canvas bank code and that bank's question number on each past test it appeared on (Fa25 = Fall 2025, Sp26 = Spring 2026). The JSON line gives the question number in each episode's TeacherCompose file, with its ID. A dash means no match exists. CHECK notes flag where this review and Test 3A differ.

WHERE TO FIND THE ANSWERS	PASSWORD
Module 5A	Module5A
5AE1	5AE1SG
5AE2	5AE2SG
Atmospheric Chart	Atm
Atmospheric Pressure exercise	AtmosSG
Ancient Atmospheres video	AASG
EvolLand exercise	EvolSG
Module 6A	Module6A
6AE1	6AE1SG
Module 10A	Module10A

The Earth's Atmosphere

1. Excluding water vapor, what four gases make up most of the atmosphere?

ANSWER Nitrogen (N_2) 78%, oxygen (O_2) 21%, argon (Ar) 0.93%, and carbon dioxide (CO_2) 0.04%.

Find answers in: Module 5A, 5AE1
QTI: T2P22 = Fa25 Test 2 #5, Sp26 Test 2 #5
JSON: 5AE1 Q2 (5AE1-P1Q2); 5AE1 Q3 (5AE1-P1Q3)

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2. Name the four layers of the atmosphere, from the ground up.

ANSWER Troposphere, stratosphere, mesosphere, thermosphere. Memory aid: "Trains Stop More Times."

Find answers in: Module 5A, 5AE1, Atmospheric Chart
QTI: —
JSON: 5AE1 Q5 (5AE1-P1Q5)

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3. In which two layers does the temperature DECREASE as you go higher, and in which two does it INCREASE?

ANSWER DECREASE: the troposphere and the mesosphere. INCREASE: the stratosphere and the thermosphere.

Find answers in: Module 5A, 5AE1, Atmospheric Chart
QTI: T2P24 = Fa25 Test 2 #2, Sp26 Test 2 #2
JSON: 5AE1 Q20 (5AE1-P4Q3); 5AE1 Q38 (5AE1-P6Q4); 5AE1 Q43 (5AE1-P6Q9)

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4. Which two gases in the stratosphere protect life from deadly UV radiation?

ANSWER Ozone (O_3) and oxygen (O_2).

Find answers in: Module 5A, 5AE1
QTI: T2P23 = Fa25 Test 2 #4, Sp26 Test 2 #4
JSON: 5AE1 Q21 (5AE1-P4Q4); 5AE1 Q22 (5AE1-P4Q5)

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5. Up to what frequency does the ionosphere reflect radio waves?

ANSWER About 30 MHz (30 megahertz).

Find answers in: Module 5A, 5AE1
QTI: T2P27 = Fa25 Test 2 #3, Sp26 Test 2 #3
JSON: 5AE1 Q29 (5AE1-P5Q4)

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6. What sphere traps the charged particles of the solar wind?

ANSWER The magnetosphere, Earth's magnetic field.

Find answers in: Module 5A, 5AE1
QTI: T2P35 = Fa25 Test 2 #1, Sp26 Test 2 #1
JSON: 5AE1 Q39 (5AE1-P6Q5)

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7. Name the six main ingredients of Earth-based weather.

ANSWER The atmosphere, water (in all three phases), the Sun, the rotation of the Earth, volcanic activity, and biological activity.

Find answers in: Module 5A, 5AE2
QTI: T2P26 = Fa25 Test 2 #6, Sp26 Test 2 #6
JSON: 5AE2 Q2 (5AE2-P1Q2); 5AE2 Q3 (5AE2-P1Q3)

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8. Name the four forms of precipitation.

ANSWER Rain, snow, sleet, and hail.

Find answers in: Module 5A, 5AE2
QTI: T2P21 = Fa25 Test 2 #7, Sp26 Test 2 #7
JSON: 5AE2 Q7 (5AE2-P2Q2)

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9. Are fog, dew, and clouds forms of precipitation?

ANSWER No. Precipitation is water falling from the sky; fog, dew, and clouds do not fall.

Find answers in: Module 5A, 5AE2
QTI: T2P21 = Fa25 Test 2 #7, Sp26 Test 2 #7
JSON: —

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10. In summer, what are cumulonimbus (storm) clouds mostly made of?

ANSWER Liquid water droplets.

Find answers in: Module 5A, 5AE2
QTI: T2P28 = Fa25 Test 2 #8, Sp26 Test 2 #8
JSON: —

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11. Name three greenhouse gases.

ANSWER Any three of: carbon dioxide, methane, water vapor, nitrous oxide, ozone. (Nitrogen and oxygen are NOT greenhouse gases.)

Find answers in: Module 5A, 5AE2
QTI: T2P25 = Fa25 Test 2 #10, Sp26 Test 2 #9
JSON: 5AE2 Q25 (5AE2-P5Q3); 5AE2 Q26 (5AE2-P5Q4)

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12. What kind of weather does high pressure bring, and what kind does low pressure bring?

ANSWER High pressure brings fair, clear weather. Low pressure brings clouds, rain, and snow.

CHECK Test 3A #2 asks whether Northern Hemisphere lows rotate clockwise (keyed False; they rotate counterclockwise). No review question covers rotation.

Find answers in: Module 5A, 5AE2
QTI: T2P30 = Sp26 Test 3 #22
JSON: 5AE2 Q29 (5AE2-P6Q2)

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Atmospheric Pressure Exercise

13. How does atmospheric pressure change as you go higher?

ANSWER It decreases, because there is less air above you.

Find answers in: Atmospheric Pressure exercise
QTI: —
JSON: ATMOS01-1; ATMOS01-2

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14. What is one Atmosphere in millibars, psi, and inches of mercury?

ANSWER 1 Atm = 1,013.25 mb = 14.7 psi = 29.92 in Hg.

Find answers in: Atmospheric Pressure exercise
QTI: T2P31 = Fa25 Test 3 #23, Sp26 Test 3 #20; T2P37 = Fa25 Test 3 #24
JSON: ATMOS01-5

Ancient Atmospheres

15. Which organisms caused the Great Oxygenation Event, and about when?

ANSWER Cyanobacteria, by photosynthesis, starting about 2.4 billion years ago.

Find answers in: Ancient Atmospheres video
QTI: T2P39 = Fa25 Test 3 #26
JSON: —

EvolLand Exercise

16. What was the Cambrian Explosion, and about when did it happen?

ANSWER A rapid diversification of complex animal life, about 540 million years ago.

Find answers in: EvolLand exercise
QTI: T2P38 = Fa25 Test 3 #25
JSON: —

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17. During the Carboniferous Period, about 360 to 300 million years ago, oxygen climbed to about what level, and what happened to insects?

ANSWER About 35% (it is about 21% today). With so much oxygen, insects grew to giant sizes; some dragonfly-like insects had wingspans of about 70 cm (over 2 feet).

CHECK The full review said oxygen peaked at about 30%; Test 3A #24 keys 35%. This edition uses 35% to match the test.

Find answers in: EvolLand exercise
QTI: —
JSON: —

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18. About how long ago did our species, Homo sapiens, evolve?

ANSWER About 300,000 years ago.

CHECK PROVISIONAL: drafted without the EvolLand exercise or its guides. Check it against EvolLand.

Find answers in: EvolLand exercise
QTI: T3A32 = Fa25 Test 4 #28
JSON: —

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The Nature of Light

19. What is optics?

ANSWER The study of light in all its forms, both visible and invisible.

Find answers in: Module 6A, 6AE1
QTI: T3A2 = Fa25 Test 3 #4, Sp26 Test 3 #3; T3A3 = Fa25 Test 3 #5, Sp26 Test 3 #4
JSON: 6AE1 (NOL) Q1 (NOL-1-01)

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20. What is the simple definition of light?

ANSWER Light is electromagnetic radiation.

Find answers in: Module 6A, 6AE1
QTI: —
JSON: 6AE1 (NOL) Q2 (NOL-1-02)

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21. Rounded, what is the speed of light in a vacuum, in miles/s and in km/s?

ANSWER About 186,000 miles/s, or about 300,000 km/s.

Find answers in: Module 6A, 6AE1
QTI: T3A9 = Fa25 Test 3 #10, Sp26 Test 3 #9
JSON: 6AE1 (NOL) Q4 (NOL-1-04)

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22. Who first used astronomy to determine the speed of light?

ANSWER Ole Rømer, in 1676, by timing Jupiter's moons.

Find answers in: Module 6A
QTI: T3A7 = Fa25 Test 3 #8, Sp26 Test 3 #7
JSON: —

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23. What do temporally coherent and spatially coherent mean? Which is laser light?

ANSWER Temporally coherent: all of one frequency (monochromatic). Spatially coherent: the wave fronts stay in step, crests lined up, across the width of the beam. Laser light is both.

Find answers in: Module 6A, 6AE1
QTI: T3A5 = Fa25 Test 3 #6, Sp26 Test 3 #5
JSON: 6AE1 (NOL) Q21 (NOL-4-05)

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24. List the electromagnetic spectrum in order of increasing frequency. What wavelengths can humans see?

ANSWER Radio/TV, microwave, infrared, visible, ultraviolet, x-ray, gamma ray. Humans see about 400 to 700 nanometers.

Find answers in: Module 6A, 6AE1
QTI: T3A2 = Fa25 Test 3 #4, Sp26 Test 3 #3
JSON: 6AE1 (NOL) Q23 (NOL-5-02); 6AE1 (NOL) Q26 (NOL-5-05); 6AE1 (NOL) Q28 (NOL-5-07)

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25. What are the three ways light interacts with matter?

ANSWER Reflection, transmission, and absorption. (A material that transmits light but scatters it is translucent.)

Find answers in: Module 6A, 6AE1
QTI: T3A11 = Fa25 Test 3 #2, Sp26 Test 3 #1
JSON: 6AE1 (NOL) Q29 (NOL-6-01); 6AE1 (NOL) Q31 (NOL-6-03)

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26. Name the five classical phenomena of light.

ANSWER Reflection, refraction, dispersion, diffraction (bending around corners), and the Doppler Effect.

Find answers in: Module 6A, 6AE1
QTI: T3A8 = Fa25 Test 3 #9, Sp26 Test 3 #8; T3A12 = Fa25 Test 3 #12, Sp26 Test 3 #11
JSON: 6AE1 (NOL) Q35 (NOL-7-03); 6AE1 (NOL) Q36 (NOL-7-04); 6AE1 (NOL) Q38 (NOL-7-06)

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27. What is the difference between specular and diffuse reflection?

ANSWER Specular reflection (a mirror) keeps the image recognizable. Diffuse reflection (paper) scatters light in random directions.

Find answers in: Module 6A
QTI: T3A6 = Fa25 Test 3 #7, Sp26 Test 3 #6
JSON: —

The Exploration of Space

28. What was Sputnik 1?

ANSWER The first artificial satellite, launched by the Soviet Union on October 4, 1957.

Find answers in: Module 10A
QTI: T4A7 = Fa25 Test 4 #4, Sp26 Test 3 #25
JSON: —

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29. What was the first satellite the United States launched successfully?

ANSWER Explorer 1 (January 31, 1958). It discovered the Van Allen Belts.

Find answers in: Module 10A
QTI: T4A7 = Fa25 Test 4 #4, Sp26 Test 3 #25
JSON: —

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30. Who was the first person to orbit the Earth?

ANSWER Yuri Gagarin, aboard Vostok 1, on April 12, 1961.

CHECK Test 3A #28 spells the answer “Gargarin.” Consider fixing it to “Gagarin.”

Find answers in: Module 10A
QTI: T4A2 = Fa25 Test 4 #2, Sp26 Test 3 #23
JSON: —

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31. Who were the first and second Americans in space?

ANSWER Alan Shepard (first) and Gus Grissom (second), both in 1961.

Find answers in: Module 10A
QTI: T4A6 = Fa25 Test 4 #29, Sp26 Test 3 #24
JSON: —

---

32. Who was the first American to orbit the Earth?

ANSWER John Glenn, on February 20, 1962, in Friendship 7.

Find answers in: Module 10A
QTI: T4A2 = Fa25 Test 4 #2, Sp26 Test 3 #23
JSON: —

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33. How many men have walked on the Moon, and where did Apollo 11 land?

ANSWER Twelve. Apollo 11 landed in the Sea of Tranquility.

CHECK Added the Apollo 11 landing site because Test 3A #29 asks it; the full review did not cover it.

Find answers in: Module 10A
QTI: T4A9 = Fa25 Test 4 #30, Sp26 Test 3 #30; T4A10 = Fa25 Test 4 #6
JSON: —

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34. Who were the first man and the first woman to fly solo nonstop across the Atlantic?

ANSWER Charles Lindbergh (May 1927, in the Spirit of St. Louis, New York to Paris) and Amelia Earhart (May 1932, Newfoundland to Northern Ireland).

CHECK Canvas T4A1 asks for "the first person to fly from the United States to Europe" (keyed Lindbergh). Alcock and Brown crossed nonstop in 1919; Lindbergh's first was solo and nonstop. Consider rewording. Earhart is not confirmed in Module 10A; check that the reading covers her, or add a line.

Find answers in: Module 10A
QTI: T4A1 = Fa25 Test 4 #1, Sp26 Test 3 #26
JSON: —

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35. Which mission sent back the first close-up images of Jupiter?

ANSWER Pioneer 10 (launched 1972), also the first spacecraft to cross the asteroid belt.

Find answers in: Module 10A
QTI: T4A3 = Fa25 Test 4 #3, Sp26 Test 3 #27
JSON: —

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36. Which mission sent back the first close-up images of Saturn?

ANSWER Pioneer 11 (launched 1973).

Find answers in: Module 10A
QTI: T4A3 = Fa25 Test 4 #3, Sp26 Test 3 #27
JSON: —

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37. What was the key first of the Viking missions?

ANSWER The first clear image from the surface of Mars (Viking Lander 1, 1976).

Find answers in: Module 10A
QTI: —
JSON: —

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38. Which spacecraft is the only one to have visited Uranus and Neptune?

ANSWER Voyager 2 (launched 1977).

Find answers in: Module 10A
QTI: —
JSON: —

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